

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
5	(a)			ZnSO ₄		1		1		
	(b)			award (1) for suitable scales on both axes award (1) for all points plotted correctly (tolerance ± 1 square) award (1) for suitable curve		3		3	2	
	(c)			curve sketched above / to the left of original curve drawn – must start from 0,0 and be horizontal at 70 cm ³ before reaching 120 s			1	1		
	(d)			rate is lower / reaction is slower (1) fewer particles in the same volume (1) less chance of (successful) collisions / fewer (successful) collisions per second / lower frequency of (successful) collisions (1)	3			3		2
	(e)			award (1) for either of following - lowers the amount of energy needed for a reaction to take place - lowers the activation energy neutral answer – catalyst not used up during reaction	1			1		1
				Question 5 total	4	4	1	9	2	3